Tanjeem Farhana Raha

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LinkedIn | GitHub | Google Scholar

Education

Bachelor of Science in Engineering in Civil Engineering — **March 2025**
Bangladesh University of Engineering and Technology (BUET) — CGPA: 3.25/4.00
Major: Transportation Engineering

Standardized Tests

GRE General Test: Quantitative 163 | Verbal 145 | Total 308
IELTS Academic: Overall Band 7.0

Research Experience

Publication

- Chowdhury, M. S. N., **Raha, T. F.**, & Adury, M. F. (2026). "Innovative Nature-Based Flood Protection Measures for Bangladesh: A Review." In *Proceedings of the 8th International Conference on Civil Engineering for Sustainable Development (ICCESD 2026)*, 5–7 February 2026, Khulna University of Engineering & Technology, Khulna, Bangladesh.

Undergraduate Thesis: "Signal System of Dhaka MRT"

- Identified the Grade of Automation (GoA) of Dhaka Metro Rail.
- Gathered information about the principal signal-system components used on the Dhaka Metro Rail main line and in the depot.
- Compared the Dhaka MRT signal system with international MRT systems in Delhi, Tokyo, Beijing, New York, and London.

Work Experience

Lecturer, Department of Civil Engineering — **October 2025–Present**
Anwer Khan Modern University
Courses taught:

- CE 2101: Mechanics of Solids 1
- CE 2201: Fluid Mechanics
- CE 3101: Structural Analysis & Design 1
- CE 3105: Structural Analysis & Design 2
- CE 2102: Mechanics of Solids 1 Sessional
- CE 3102: Structural Analysis & Design Sessional

Technical Skills

Programming and data analysis: C++, MATLAB, Python, Scikit-learn, machine-learning regression and classification, Pandas, NumPy, Matplotlib
Structural analysis: ETABS, SAP2000
Building layout and design: AutoCAD, SketchUp
Traffic simulation: Vissim
Other: Microsoft 365

Projects

Geospatial Analysis Project — GitHub Repository
"Khulna Coastal Road Flood-Exposure Analysis"

- Built a reproducible Python/GIS pipeline combining geoBoundaries, OpenStreetMap roads, and WRI Aqueduct Floods v2 rasters to assess Khulna's road exposure across 15 overlays (three 100-year scenarios × five depth thresholds), exporting validated maps and road-class summaries.

Machine Learning Project — GitHub Repository
"I-94 Hourly Traffic-Volume Prediction"

- Built an end-to-end Python/scikit-learn pipeline using 48,204 UCI observations to clean data, engineer temporal and weather features, and predict westbound hourly I-94 traffic volume.
- Compared baseline and regression models using chronological validation; the selected model achieved an MAE of 252.2 vehicles/hour and an R^2 of 0.941 on the 2018 holdout.

Capstone Project
“Rehabilitation of Malibagh–Khilgaon Slum Dwellers”

- Developed a proposal for construction of a 13-story building for slum dwellers.
- Performed ground engineering and foundation modeling.
- Conducted environmental and social impact assessment; prepared an Environmental Management Plan (EMP), plumbing design, and Traffic Impact Assessment (TIA).
- Completed a feasibility study, cost estimation, baseline studies and surveys, and market-demand analysis.

MATLAB Project
“Reinforcement and Cost Estimation of Superstructure”

- Estimated construction cost with MATLAB, calculated the total quantity of reinforcement in slabs, beams, and columns, and estimated the total cost for the required quantity.

Vissim Project
“Traffic Simulation at Shahbagh Intersection and Comparing Capacities for Different Car-Following Parameters”

- Simulated traffic at Shahbagh Intersection and evaluated capacity with the Wiedemann 74 model by varying average standstill distance and the additive and multiplicative parts of safety distance.

Public Transportation Project
“Operational Diagnosis and Service Improvement Strategies for the VIP 27 Bus Route (Azimpur to Gazipur)”

- Evaluated the present performance of the VIP 27 bus, identified root causes of current problems, collected data, compared the service with Tehran BRT, developed practical strategies, and estimated their potential benefits.

Term Project
Critical Review of “Greater Dhaka Sustainable Urban Development Project BRT (Gazipur to Dhaka)”

- Reviewed socioeconomic study frameworks, impact identification, mitigation measures, Environmental Impact Assessment (EIA), and the project’s implementation framework.

Urban Development Project
“Envisioning a More Liveable Dhaka”

- Analyzed urban challenges in Dhaka, proposed actionable and sustainable solutions to improve liveability, and synthesized the project into a visual poster presentation.

Scholarships and Awards

- **Scholarship for Descendants of Freedom Fighters, Government of India**
  - Higher Secondary student, Holy Cross College (2018)
  - Undergraduate student, BUET (2021)
- **General Scholarship in Secondary School Certificate** (2017)
- **Talent Pool Scholarship in Higher Secondary Certificate** (2019)

Seminars & Workshops

- Attended “Innovative Nature-Based Technologies for Slope Stabilization, Land Reclamation and Soil Remediation,” presented by Prof. Dr. Mohammad Shariful Islam.
- Attended “Climate Smart Bio-engineered Slope Protection Using Vetiver Grass: Scopes for Local Government Initiatives for Climate Change (LoGIC),” presented by Prof. Dr. Mohammad Shariful Islam.
- Attended “Our Freshwater Challenges: Nature Based Solutions,” presented by Retd. Dr. Md. Mujibur Rahman.
- Attended “Introduction to Efficient and Sustainable Transport System,” presented by Prof. Dr. Md. Shamsul Hoque.
- Participated in the “Women in Engineering” workshop at the 16th Inter-College Science Festival, Holy Cross College.

Extra-Curricular Activities

- Participated in “Welcoming Bangla New Year 1421 in 1000 Voices, 2014,” organized by Channel I.
- Executive Choreographer, BUET Dance Club.
- Member, Kantho BUET.
- Member, BUET Radio Club.
- Volunteer, Ceismic Civil Fest.
- Participated in Intra-BUET Online Dance Fair—Groove On 2021.
- Volunteer and performer, 6th Inter-University Dance Festival, BUET.

References

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